Notes from reference group meeting

- Example reports in prep.
- Design vs. Analysis vs. Interpretation/Presentation
- Additional reading

Processing and Analysis of Biological Data

BIOS15 2025

Lecture 10.
Causal inference

$$y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i$$

Øystein H. Opedal

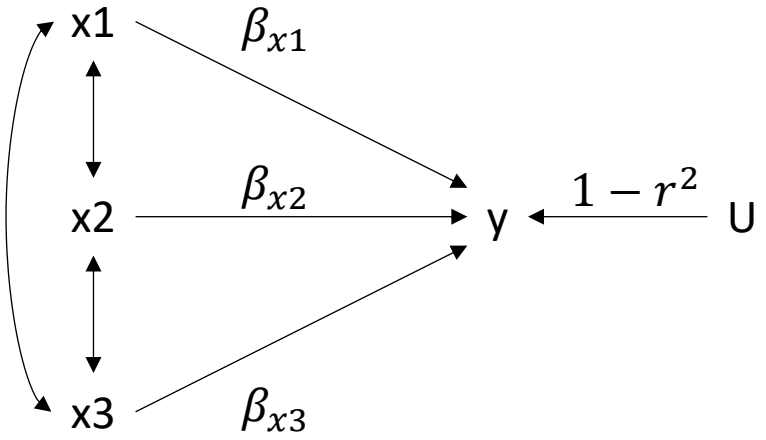
Cause and correlation

- Randomized experiments provide a powerful approach to inferring causal relations
- Sometimes, however, we can infer causality without experiments
- Doing so requires a deep understanding of the problem at hand
- Sewall Wright developed **path analysis** for this purpose

The method depends on the combination of knowledge of the degrees of correlation among the variables in a system with such knowledge as may be possessed of the causal relations. In cases in which the causal relations are uncertain the method can be used to find the logical consequences of any particular hypothesis in regard to them. (Wright 1921)

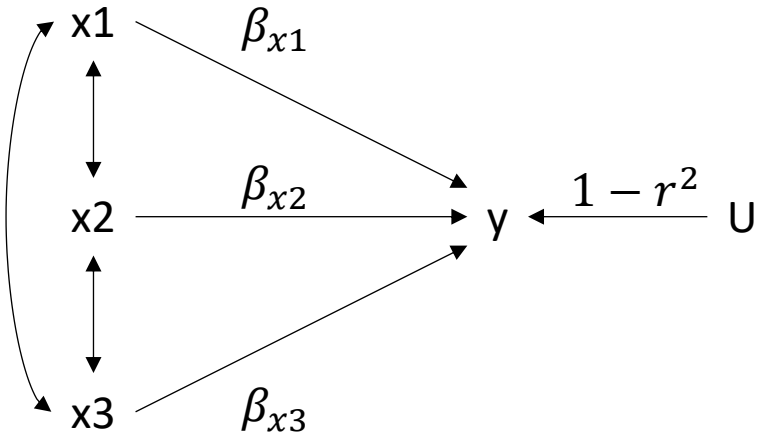
Path analysis

- A multiple-regression analysis can be seen as a simple path analysis, and we have therefore already seen a simple path diagram
- The path diagram (directed graph) can be seen as a graphical illustration of a **causal hypothesis**, but also as an **empirical roadmap** and as a **statistical model**



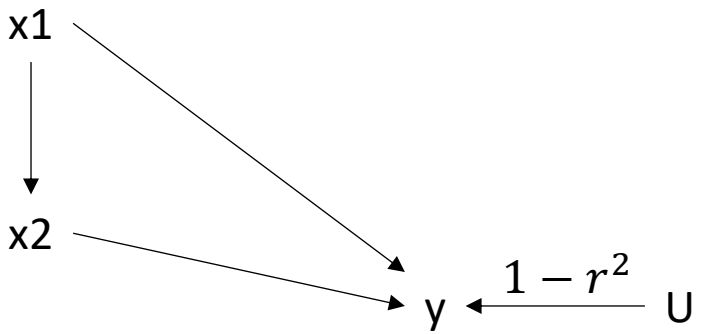
Path analysis

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- Drawing the path diagram (causal diagram) is a critical part of the analysis. Estimating the parameters is merely the last step
- In Wrightian path analysis, we z-transform all variables (also the response) to get standardized coefficient («path coefficients»).
- «U» indicates the unexplained variance



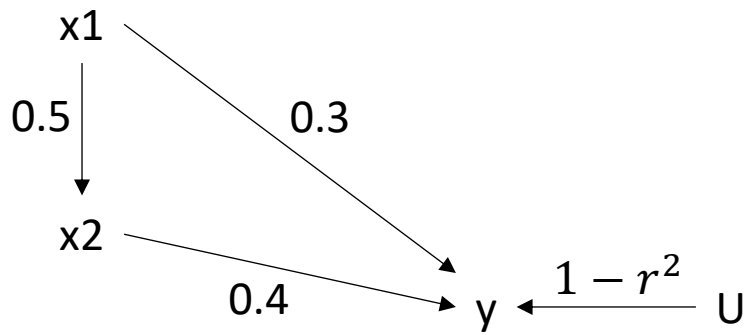
Path analysis: direct and indirect effects

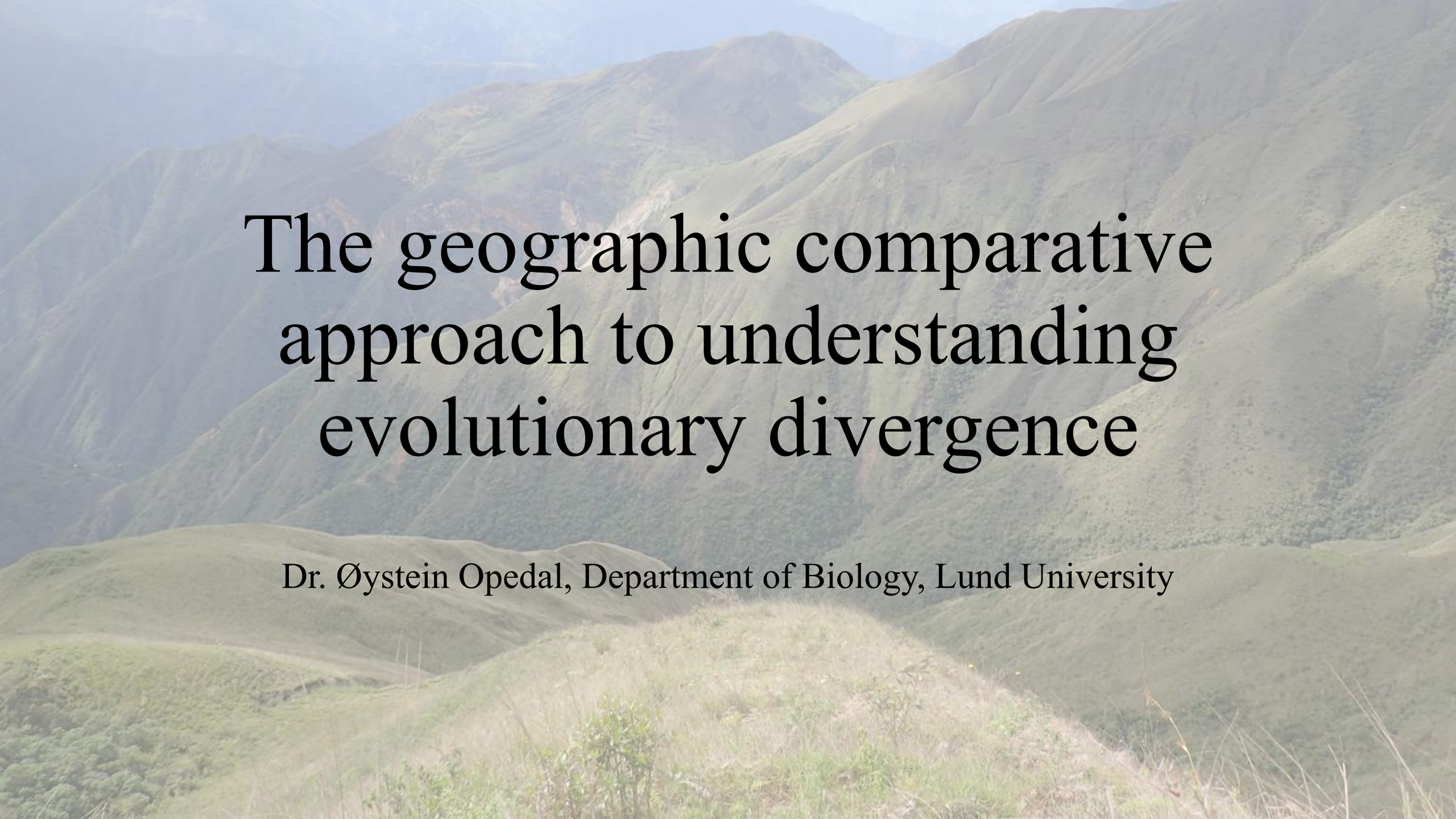
- Single-headed arrows indicate direct (causal) effects, double-headed arrows indicate (non-causal) correlations
- **Endogeneous** variables have arrows coming into them, **exogeneous** variables have only arrows going out
- Here, x1 has both an direct effect on y, and an indirect effect via x2
- To get the net effect of x1 on y, we sum both paths (x1 -> y + x1 -> x2 -> y)



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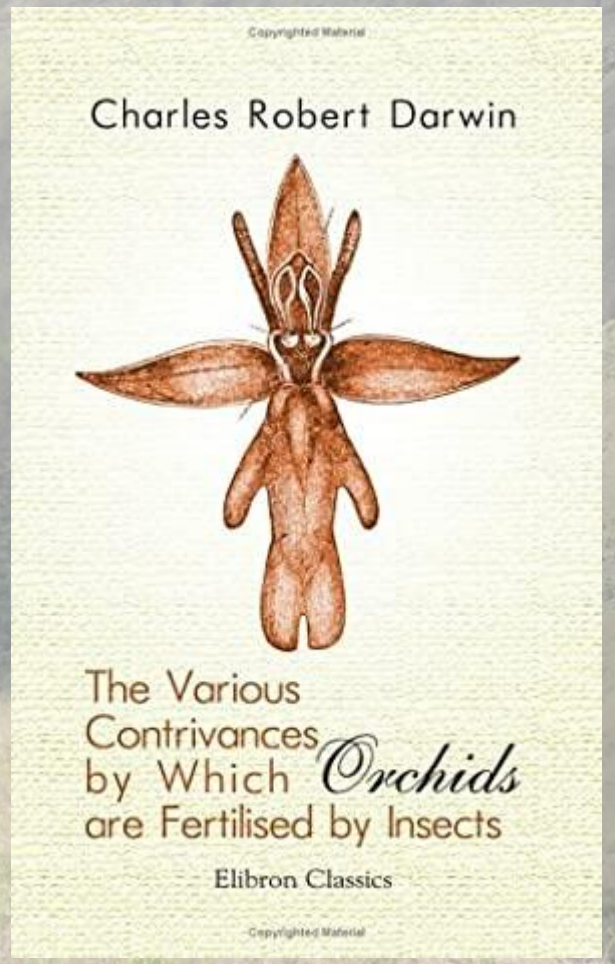


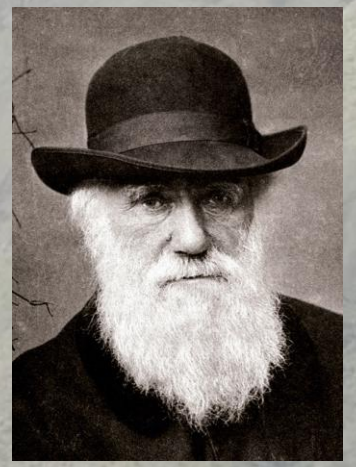
The geographic comparative approach to understanding evolutionary divergence

Dr. Øystein Opedal, Department of Biology, Lund University

The study of floral evolution: Darwin's 'flank movement on the enemy'

- 'no one else has perceived that my chief interest in my orchid book, has been that it was a "flank movement" on the enemy' (Darwin to Asa Gray, July 23rd 1862)
- 'If you grant an intelligent designer anywhere in Nature, you may be confident that he has had something to do with the "contrivances" in your Orchids.' (Gray to Darwin, July 2nd 1862)





The ‘Grant-Stebbins model’

- Plant speciation often results from adaptation to variable ‘pollination climates’:
- Flowers and plant-pollinator interactions play key roles in plant speciation
- Adaptive radiation in animal-pollinated plants is often related to traits affecting pollination, as reflected by diagnostic characters in taxonomy
- **Predicts covariation between floral phenotypes and pollinator assemblages**



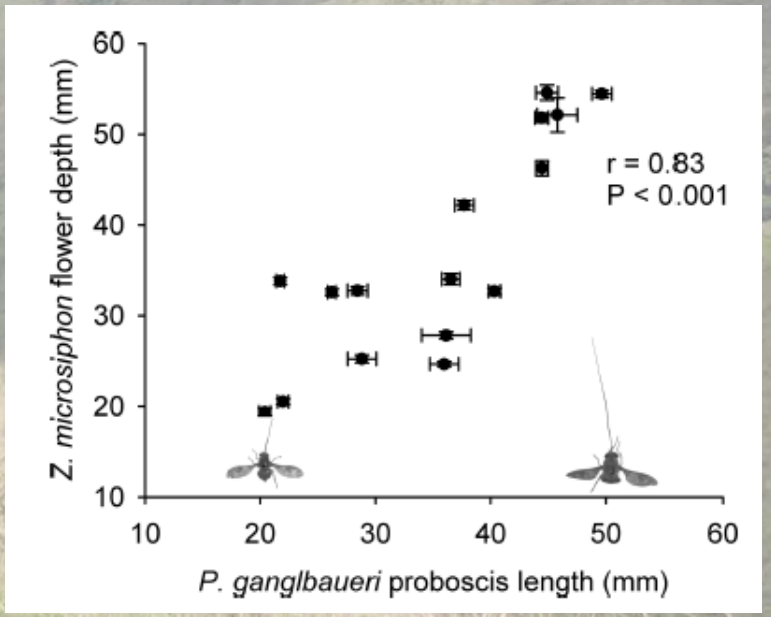


e.g. Grant 1949, Grant and Grant 1965, Stebbins 1970, 1974, Johnson 2006

The geographic comparative approach

- By comparing populations occurring under different environmental conditions, we can test hypotheses about phenotypic adaptation
- **When combined with prior hypotheses about cause and effect, observational data can be used to assess causality**

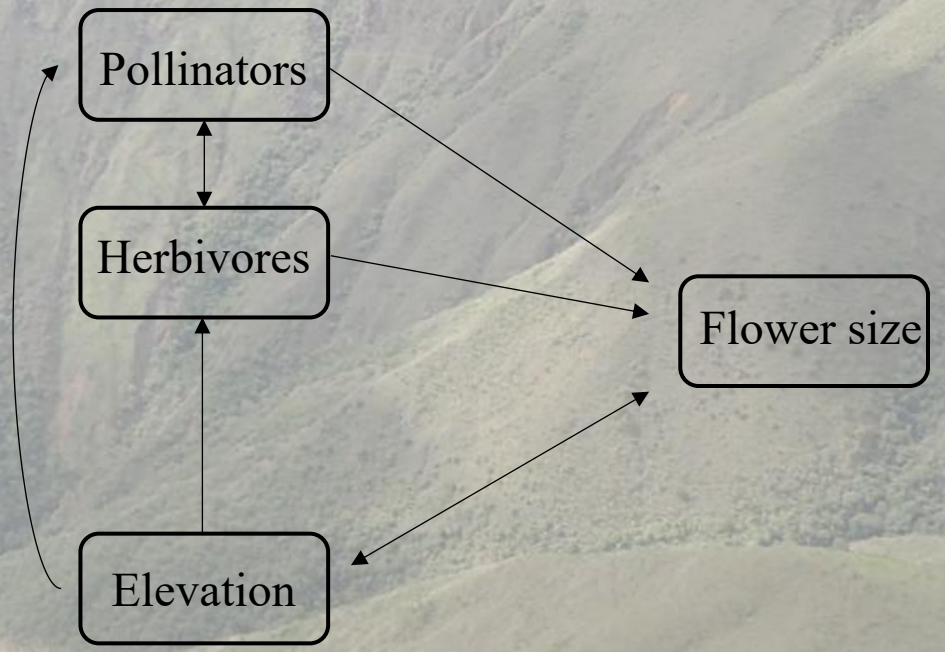




Anderson & Johnson 2008 Evolution

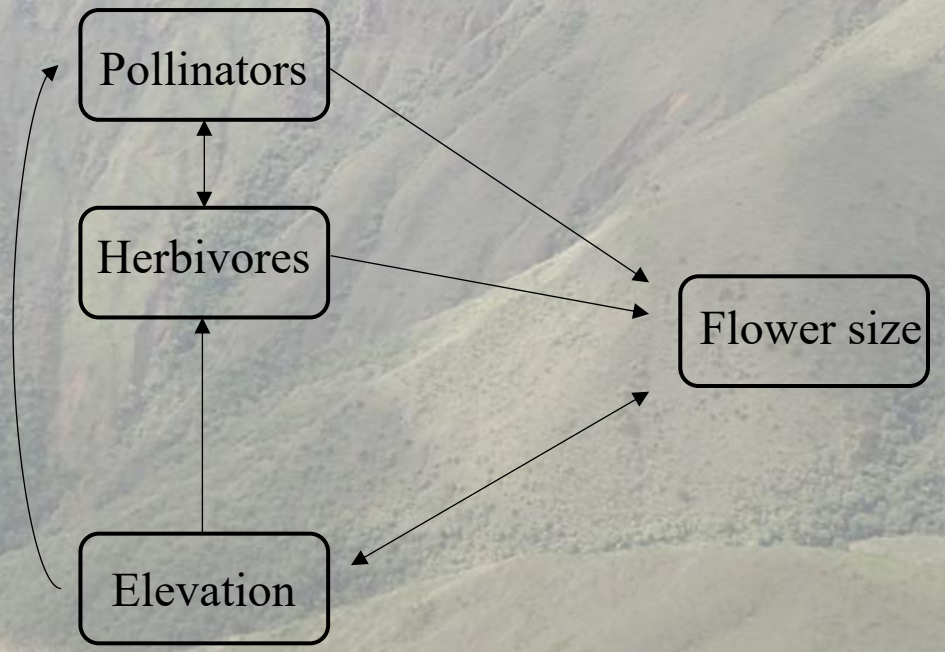
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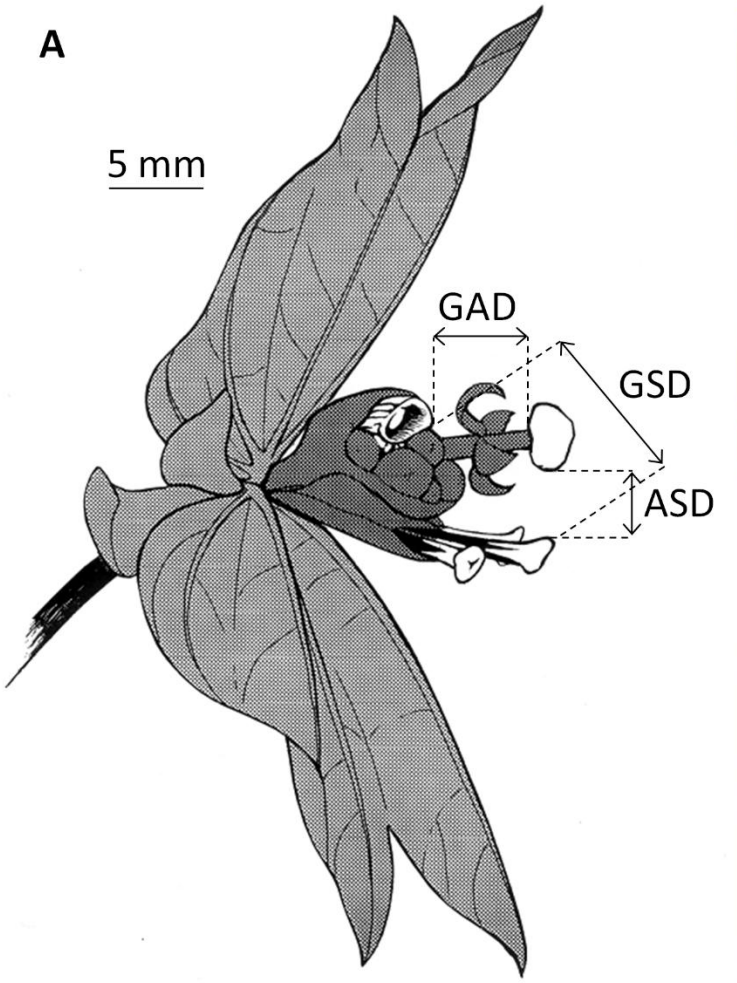


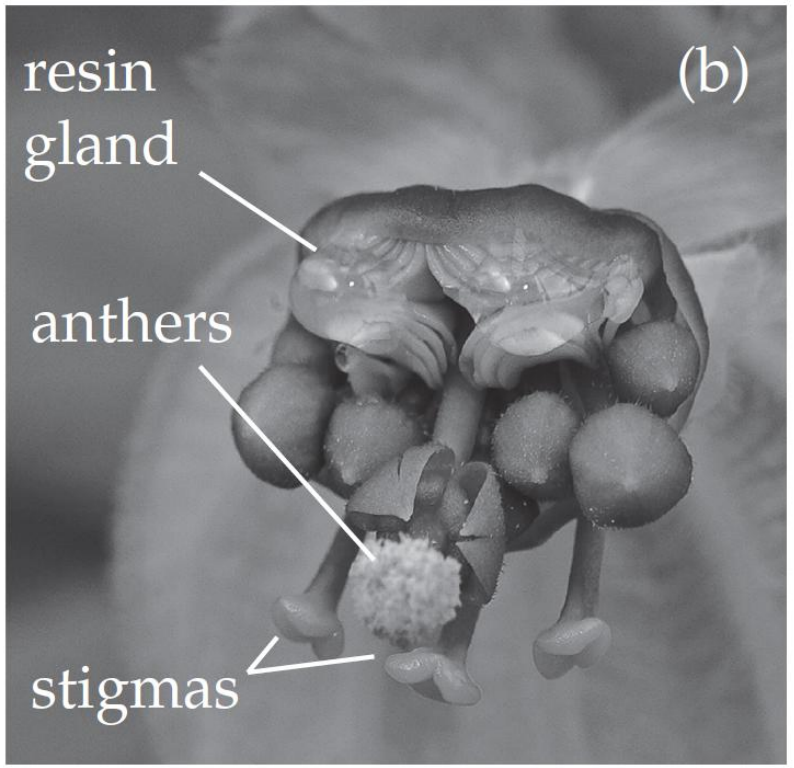
Model system: *Dalechampia L.* (Euphorbiaceae)

- Pantropical genus with majority of species occurring in the neotropics
- Long-term model of pollinator-mediated floral evolution
- Most neotropical species pollinated by resin-collecting female bees, though 3 clades have shifted to pollination by fragrance-collecting male euglossine bees, and several to pollen rewards (including buzz pollination in Madagascar)



Female phase

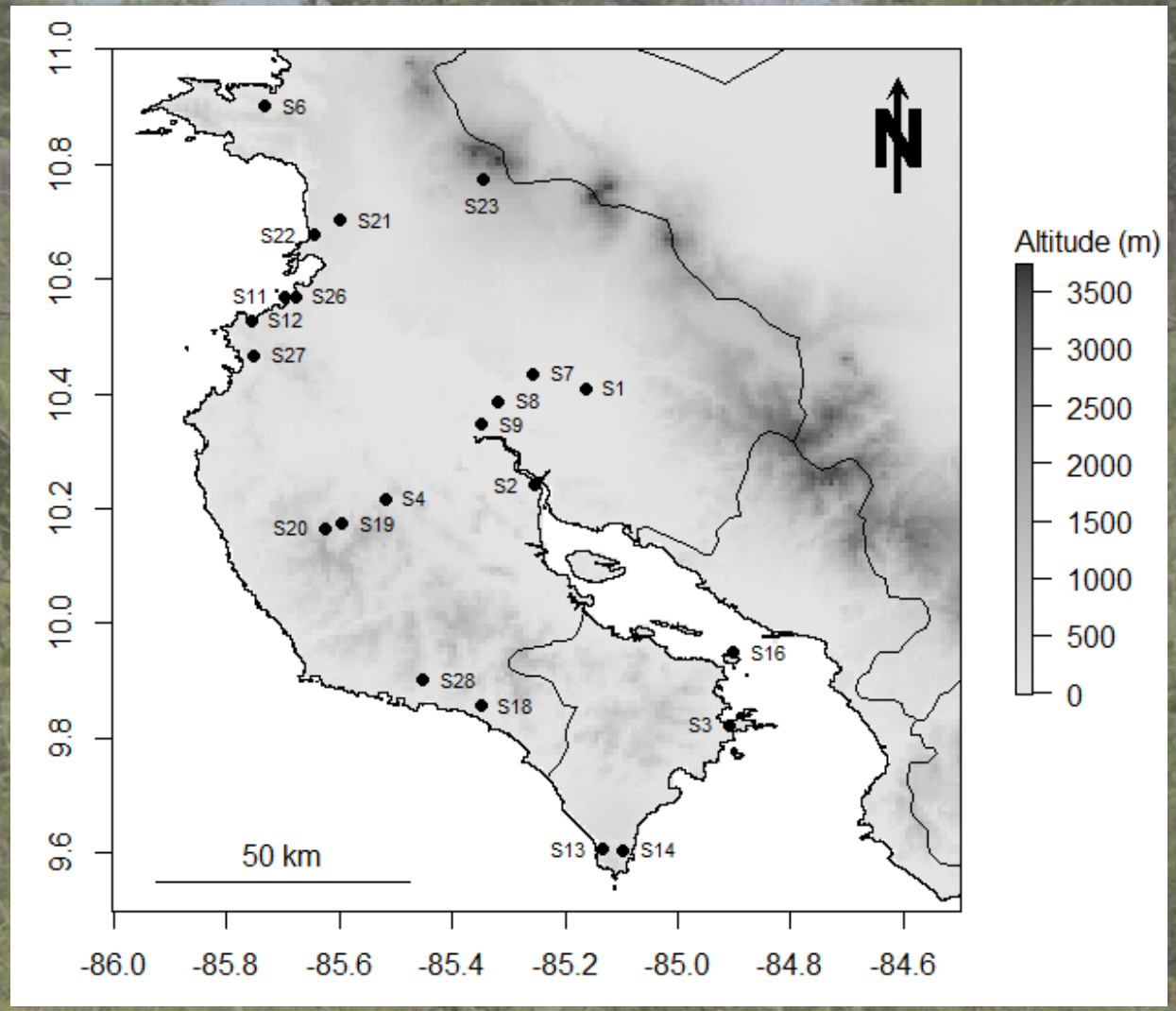




Bisexual phase

The Costa Rica study system

- ~22 populations



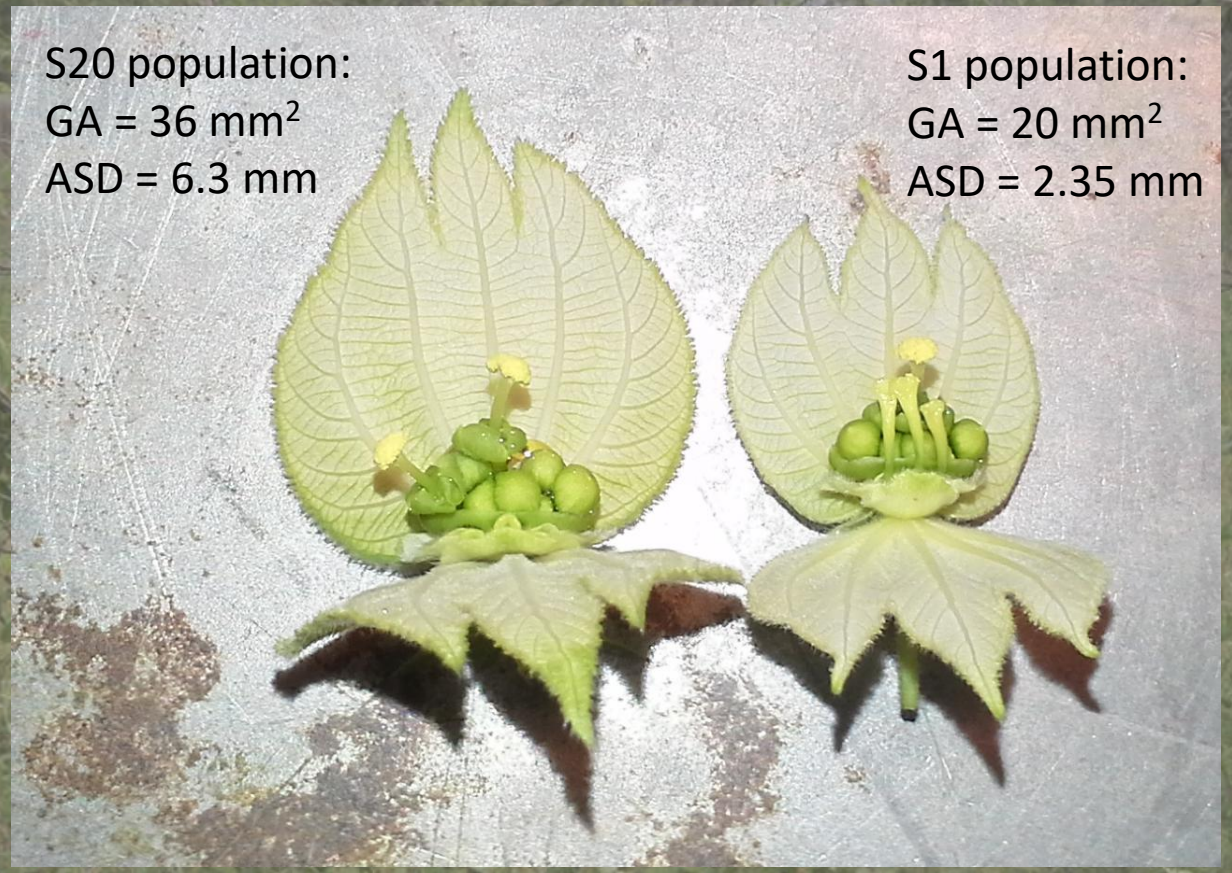
The Costa Rica study system

- ~22 populations
- Field data on pollinators and seed predators



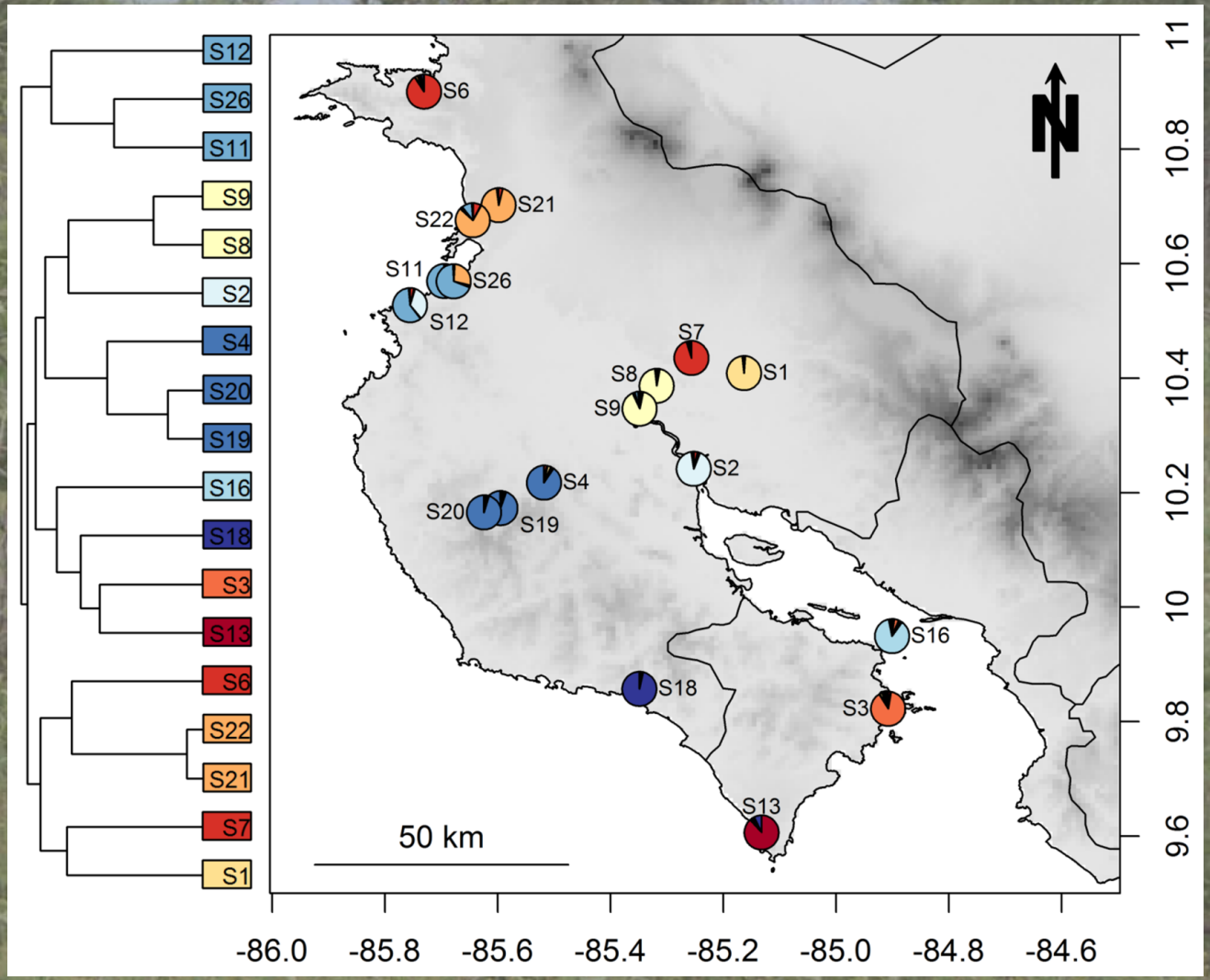
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The Costa Rica study system

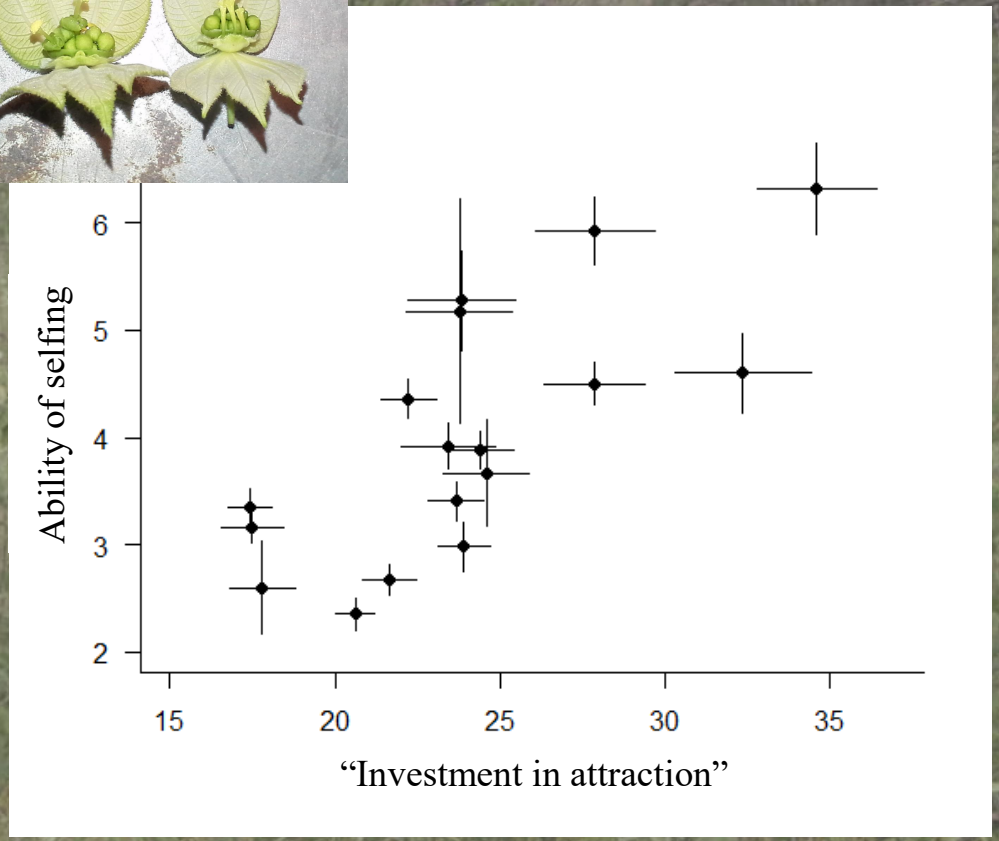
- ~22 populations
- Field data on pollinators and seed predators
- Phenotypic data (field + greenhouse)
- Genetic (microsatellite) data



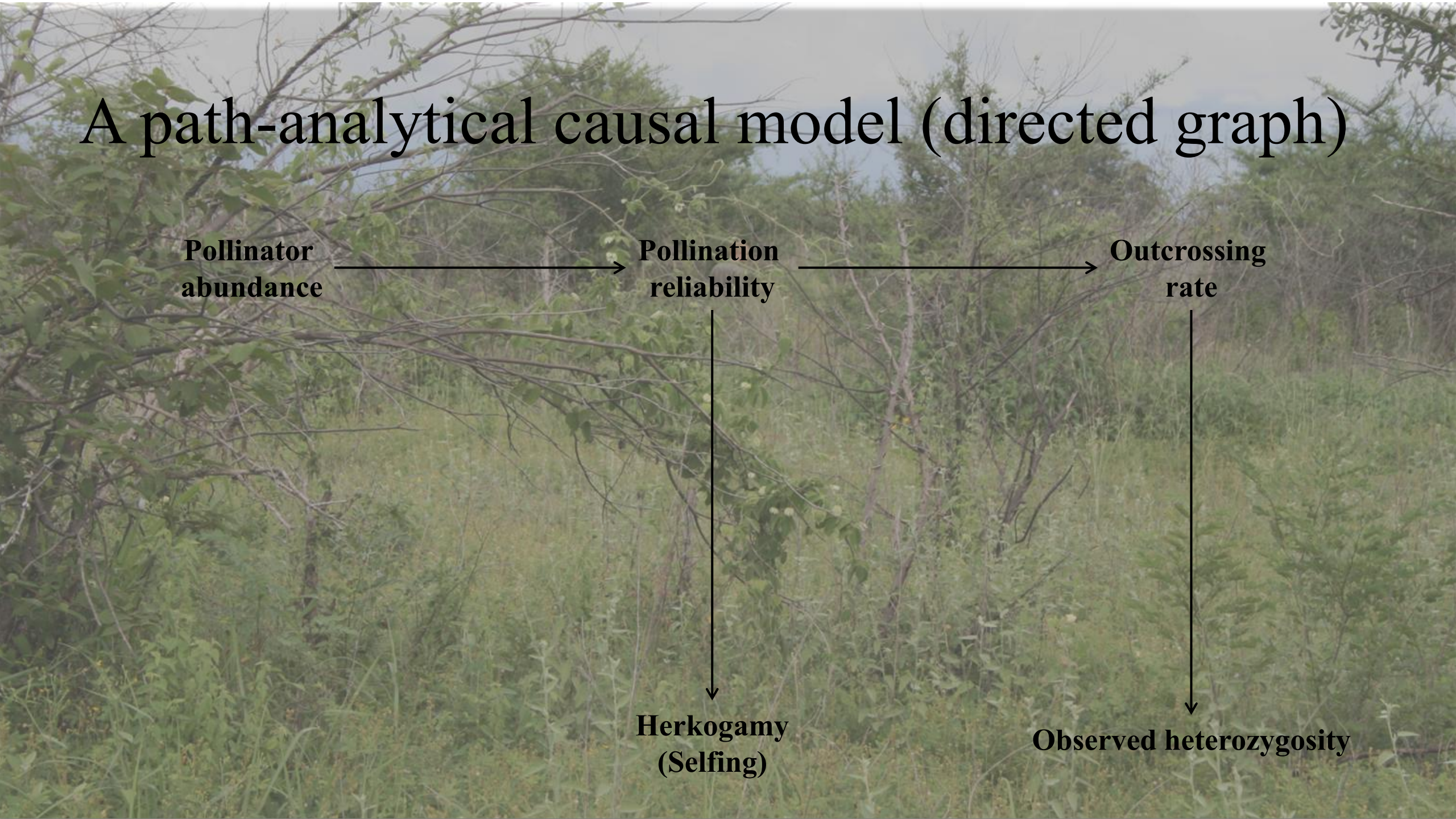
A hypothesized mating-system gradient

- Traits covary along an axis of increasing trait values
- We can hypothesize that this represents a gradient from selfing to (more) outcrossing
- Implies *selective covariance*, here likely due to a joint selective agent (pollinators).





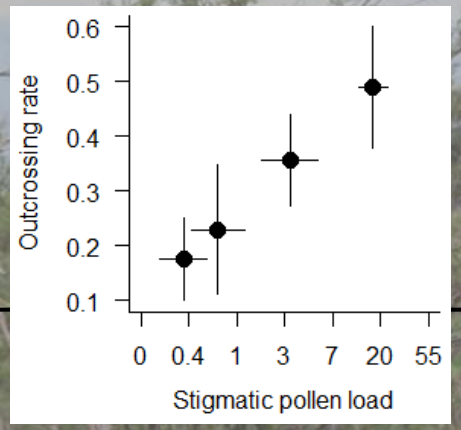
A path-analytical causal model (directed graph)



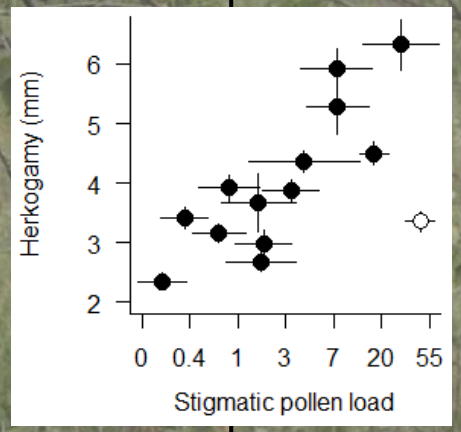
Pollinator abundance



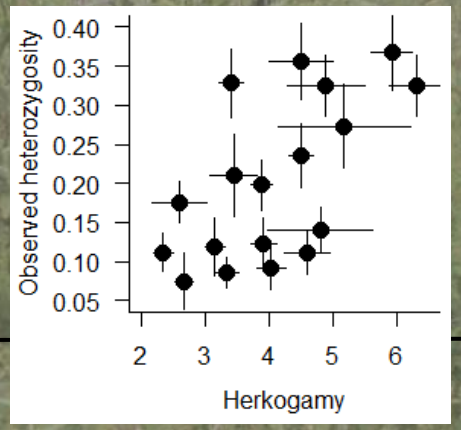
Pollination reliability



Outcrossing rate



Herkogamy (Selfing)



Observed heterozygosity

Euglossine bees

- Tribe Euglossini (Apidae)
- *Eulaema*, *Eufriesea* & *Euglossa*
- *Exaerete* & *Aglae* (cleptoparasites)
- ~230 species, all occur in tropical America. Typically one forth of bee species in tropical forests
- Mostly solitary, sometimes colonial and primitively eusocial



Female resource needs

- Nectar
- Pollen
- Resin for nest construction



Clusia sp.
(Clusiaceae)



Euglossa dilemma
Dalechampia scandens



Eufriesea sp.
Clusia sp.

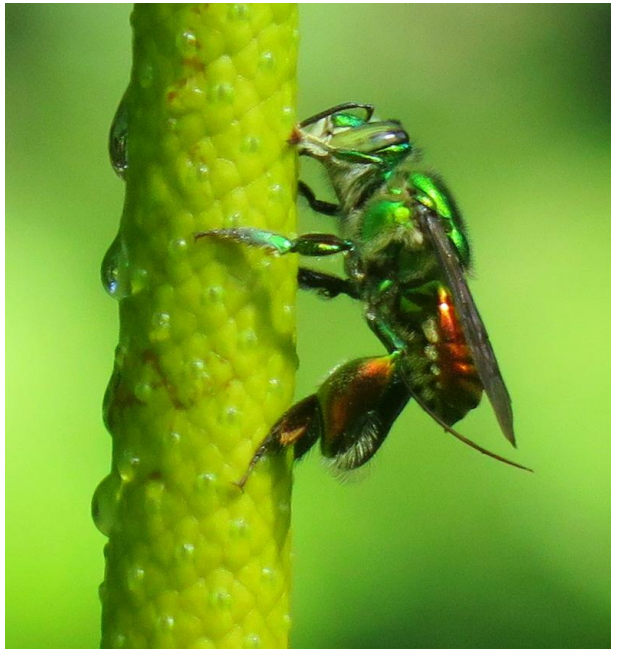


Male resource needs

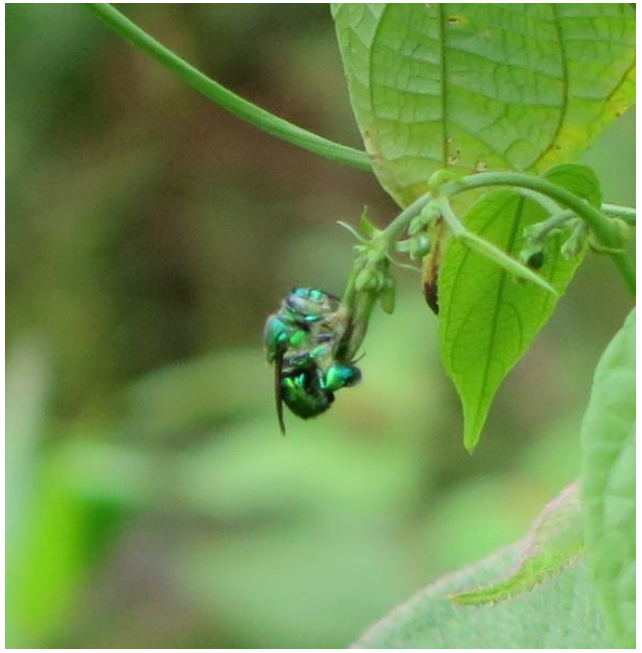
- Nectar
- Fragrances for courtship displays



Euglossa hansoni on *Gongora*
Photo by S. Ramírez



Euglossa cf. *ignita* on *Anthurium*



Euglossa modestior on
Dalechampia gentryi

Baiting euglossines

- Simple way to estimate density and diversity of male bees
- No info about female bees (?)

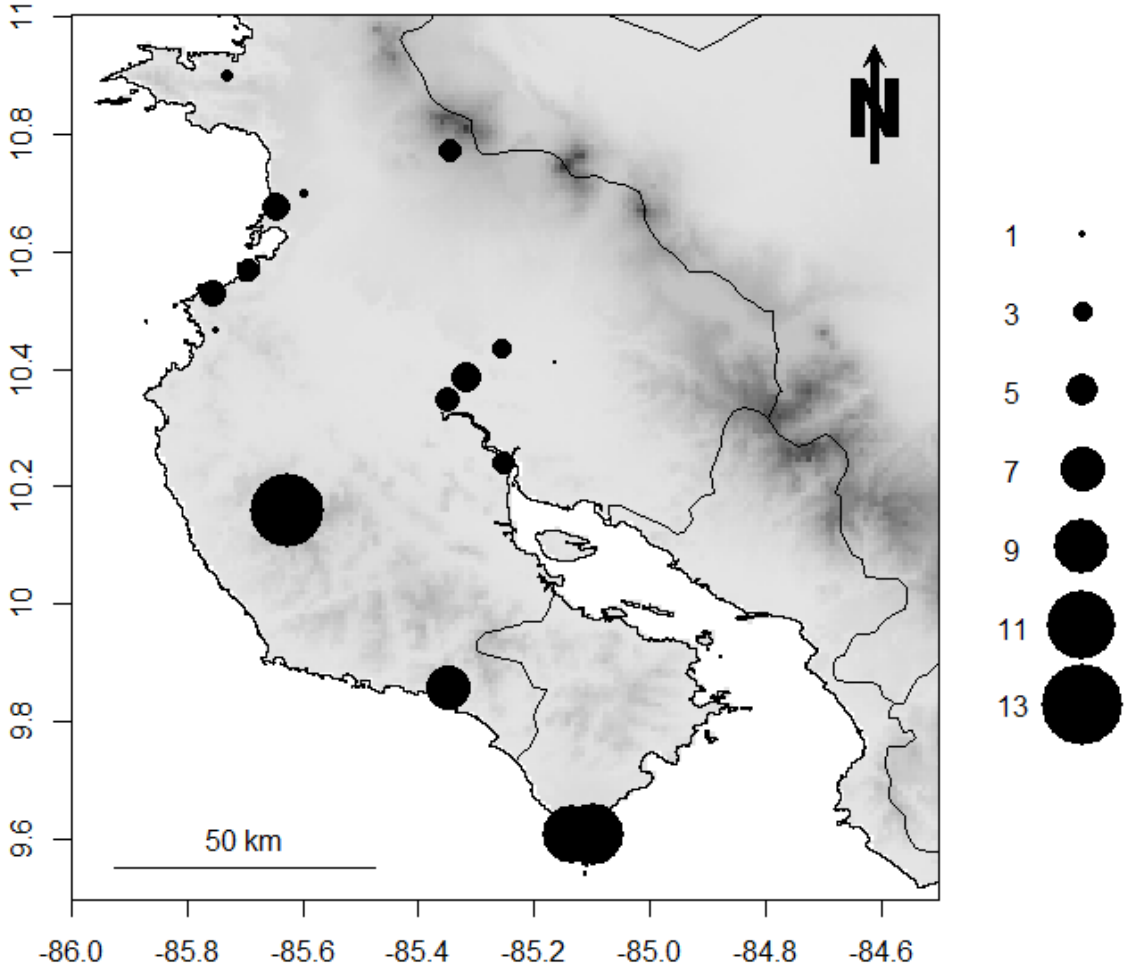


Male euglossines in Guanacaste

Male *Euglossa dilemma* on bait:



N = 64 baiting sessions,
482 bees recorded on Eugenol
and Methyl cinnamate

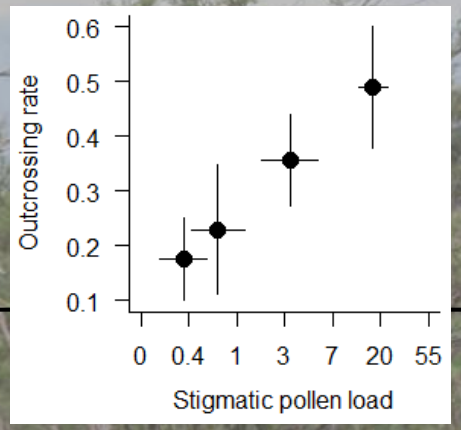


Opedal, Albertsen & Armbruster, unpublished.

Pollinator abundance



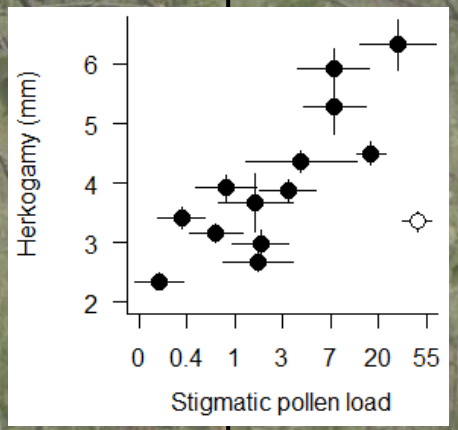
Pollination reliability



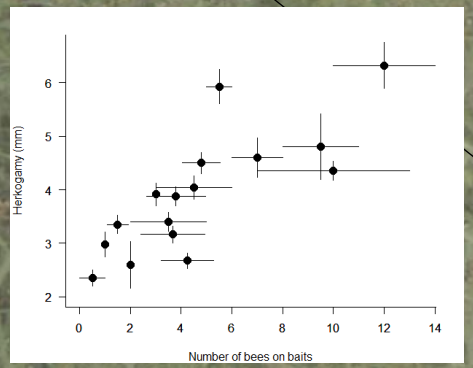
Outcrossing rate



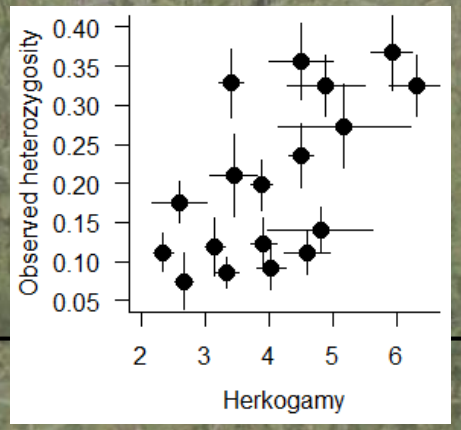
r = 0.46



r = 0.79



Herkogamy



Observed heterozygosity

More than just pollinators

- *Dalechampia* seeds are eaten by the larvae of Baridinae seed weevils

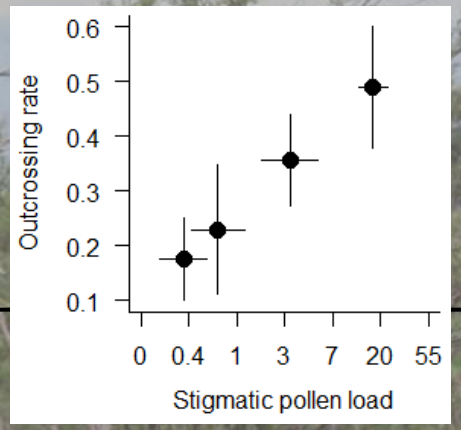




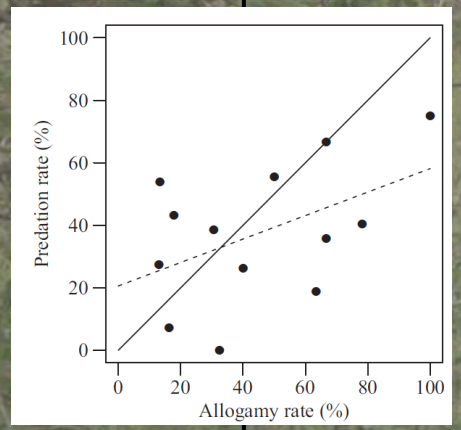
Pollinator abundance

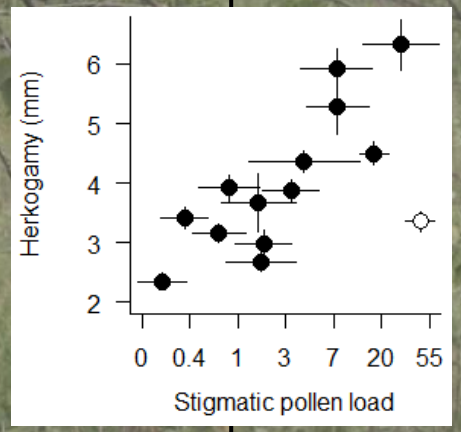


Pollination reliability



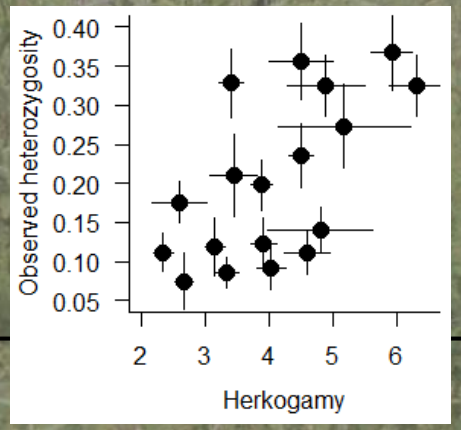
Outcrossing rate





Weevil abundance

Herkogamy



Observed heterozygosity





Mating system

Predation rate

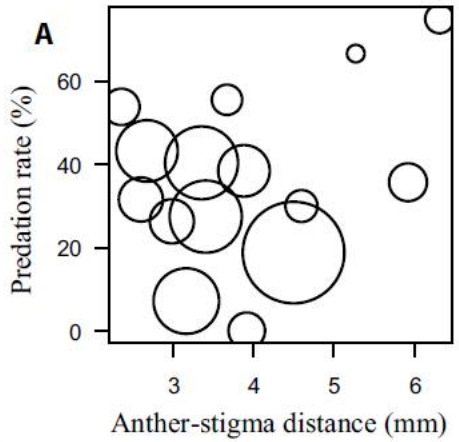
0.69
(0.18, 1.20)

0.34
(-0.21, 0.73)

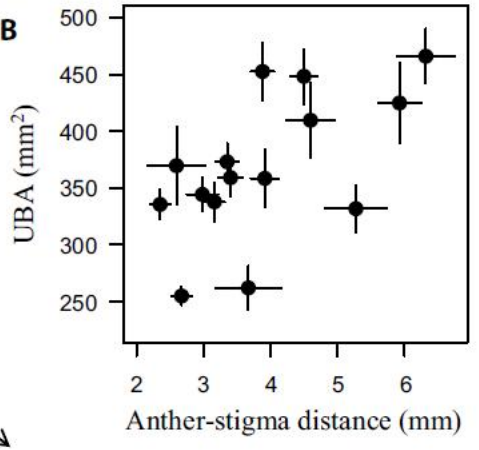
-0.27
(-0.78, 0.24)

UBA ← U
0.76

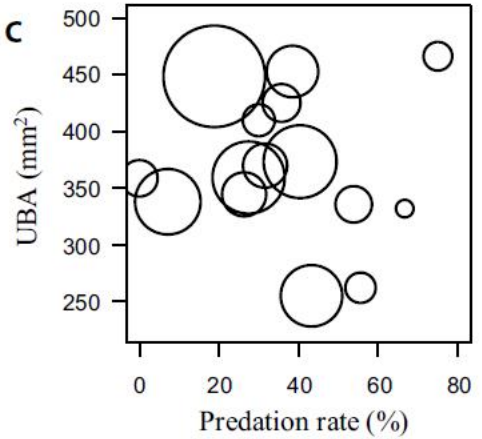
A



B



C



Opedal et al. 2019 AJB

The ‘Speciation, adaptation and coevolution’ (SPACE) research group

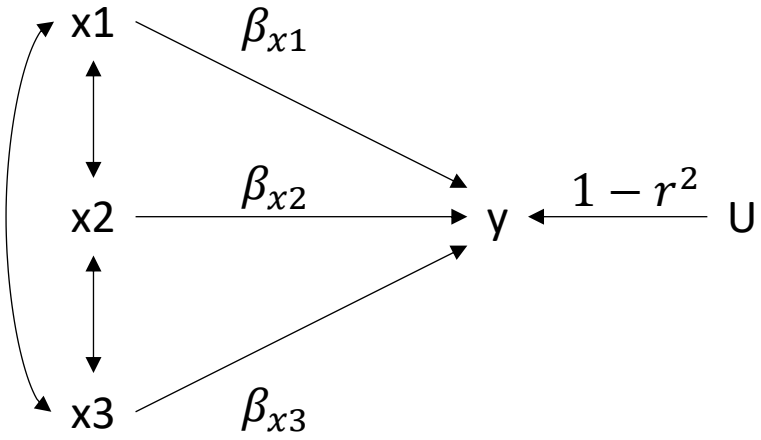
- We study evolutionary adaptation across the landscape
- Focus on plant-insect interactions
- (Plant) field systems in Sweden, Italy, California, Costa Rica, South America
- Get in touch for project opportunities!



Senior PIs: Magne Friberg, Anna Runemark, Øystein Opedal

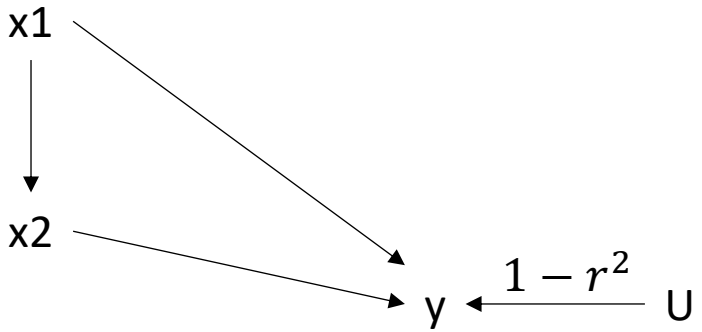
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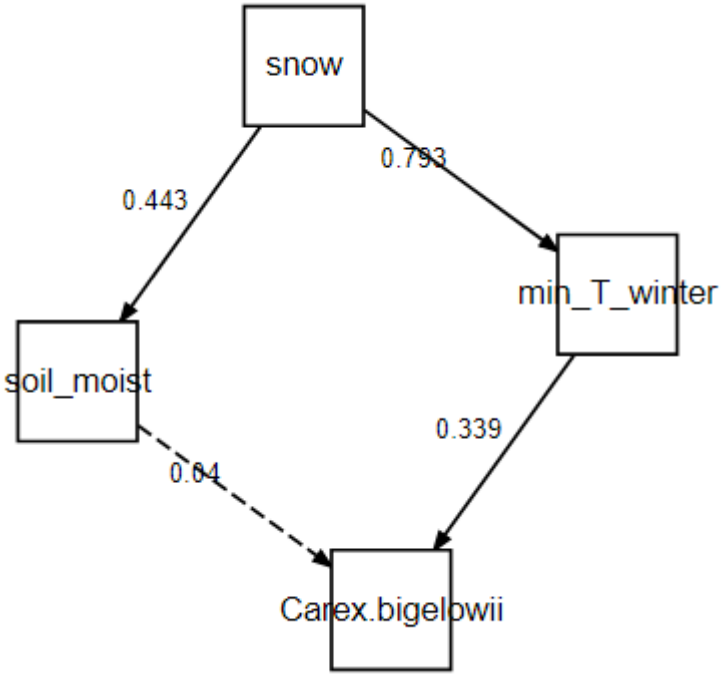
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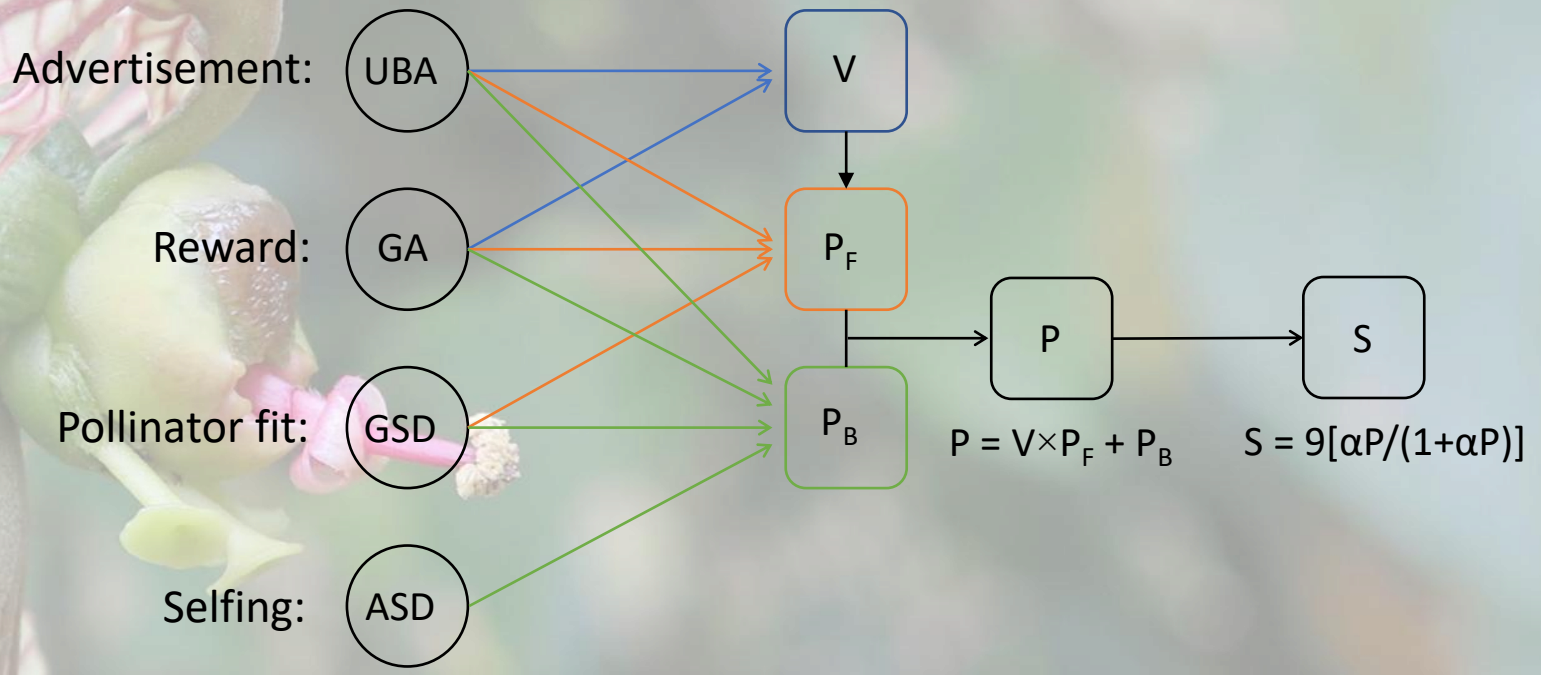


Structural equation modelling

- Structural equation modelling is a further development of path analysis in which we fit the entire model in one go
- We can test the entire model for deviation from the data to evaluate overall support
- Many extensions including unmeasured “latent” variables
- Also posible to fit individual paths as GLMs



Dalechampia fitness function



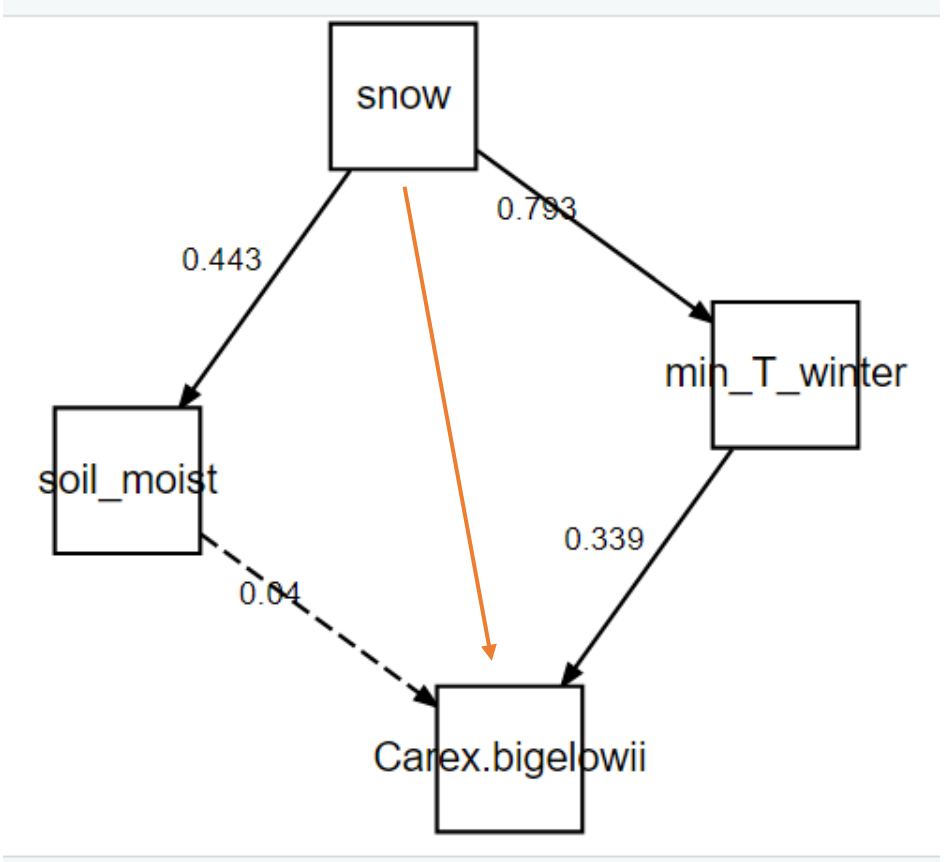
Visitation (V) = f(Advertisement, Reward)

Female-phase pollen (P_F) = f(Advertisement, Reward, Fit)

Bisexual-phase pollen (P_B) = f(Advertisement, Reward, Fit, Herkogamy)

Third gen SEM: piecewiseSEM

- Directed separation: are variables conditionally independent?
- In this model, we make the claim that *Carex* abundance is independent of variation in snow cover (the red arrow).
- Can test this by asking if snow indeed explains little additional variation while accounting for winter temperature and soil moisture.



Piecewise SEM in R

- With the piecewiseSEM package, we simply provide a list of (generalized) linear models representing the structural equation model

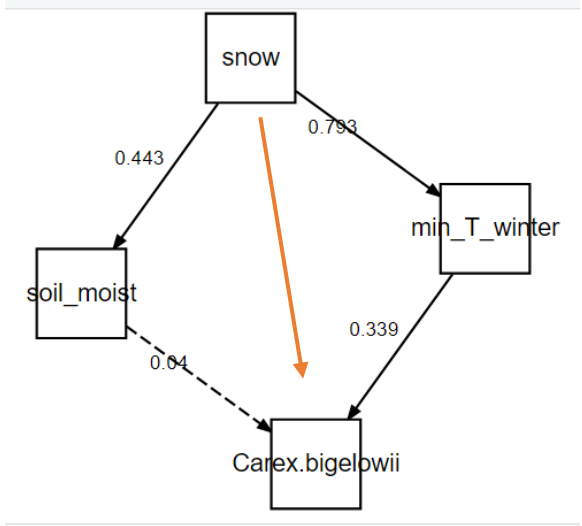
```
library(piecewiseSEM)

m2 = psem(lm(soil_moist~snow, data=plants),
          lm(min_T_winter~snow, data=plants),
          lm(Carex.bigelowii~min_T_winter+soil_moist, data=plants),
          data=plants)
```

Piecewise SEM in R

```
##
## Structural Equation Model of m2
##
## Call:
##   soil_moist ~ snow
##   min_T_winter ~ snow
##   Carex.bigelowii ~ min_T_winter + soil_moist
##
##     AIC
##  683.464
##
## ---
## Tests of directed separation:
##
##                    Independ.Claim Test.Type DF Crit.Value P.Value
##      Carex.bigelowii ~ snow + ...      coef 89     1.0875  0.2797
##   min_T_winter ~ soil_moist + ...      coef 90     1.9656  0.0524
##
## --
## Global goodness-of-fit:
##
## Chi-Squared = 5.137 with P-value = 0.077 and on 2 degrees of freedom
## Fisher's C = 8.445 with P-value = 0.077 and on 4 degrees of freedom
##
## ---
```

```
## Coefficients:
##
##          Response    Predictor Estimate Std.Error DF Crit.Value P.Value
##        soil_moist         snow   0.4434    0.0940 91     4.7188  0.0000
##      min_T_winter         snow   0.7928    0.0639 91    12.4082  0.0000
##   Carex.bigelowii min_T_winter   0.3389    0.1110 90     3.0546  0.0030
##   Carex.bigelowii   soil_moist   0.0399    0.1110 90     0.3592  0.7203
##   Std.Estimate
##         0.4434 ***
##         0.7928 ***
##         0.3389  **
##         0.0399
##
##   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05
##
## ---
## Individual R-squared:
##
##          Response method R.squared
##        soil_moist   none      0.20
##      min_T_winter   none      0.63
##   Carex.bigelowii   none      0.13
```



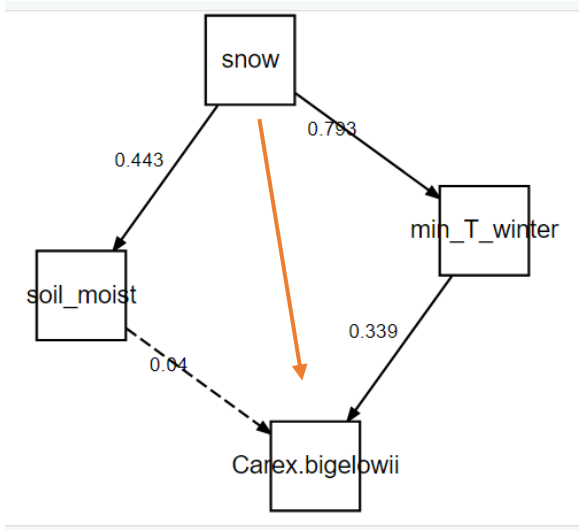
Piecewise SEM in R

- To represent an untested correlation (doubleheaded arrow), we can use the %--% operator

```
m2b = psem(lm(soil_moist~snow, data=plants),
           lm(min_T_winter~snow, data=plants),
           lm(Carex.bigelowii~min_T_winter+soil_moist, data=plants),
           min_T_winter %~~% soil_moist,
           data=plants)
```

Piecewise SEM in R

```
## Structural Equation Model of m2b
##
## Call:
##   soil_moist ~ snow
##   min_T_winter ~ snow
##   Carex.bigelowii ~ min_T_winter + soil_moist
##   min_T_winter ~~ soil_moist
##
##     AIC
##  683.464
##
## ---
## Tests of directed separation:
##
##                 Independ.Claim Test.Type DF Crit.Value P.Value
##   Carex.bigelowii ~ snow + ...      coef 89     1.0875  0.2797
##
## --
## Global goodness-of-fit:
##
## Chi-Squared = 1.228 with P-value = 0.268 and on 1 degrees of freedom
## Fisher's C = 2.548 with P-value = 0.28 and on 2 degrees of freedom
##
```



SEM model selection

- Likelihoods, and hence AIC values, are additive. piecewiseSEM also provides the AIC for the entire model, and we can use this to compare competing models or compare to a null model

```
AIC(m2b, m3)
```

```
##       AIC  K  n
## 1 683.464 10 93
## 2 681.598  9 93
```

Intro to mid-term exercise

- You will receive a dataset, formulate your own research question and design your own quantitative analyses
- The report (deadline Friday at 1600) is maximum 4 pages, and should contain a brief background, **analysis methods**, **results** (incl. relevant tables/figures, and conclusions (no Discussion). Practise for the exam:
  a. Formulation of research question(s) (10%)
  b. Choice, justification and presentation of the analysis methods (20%)
  c. Presentation of results in text (20%)
  d. Presentation of results in figures/tables (20%)
  e. Interpretation/conclusions (20%)
  f. Clarity of analysis code (10%)

SEM in R

- The fitted SEM gives the path coefficients, and also an overall test for the goodness of fit of the model
- NB: A low p-value indicates a rather **poor** fit of the model

```
## lavaan 0.6-12 ended normally after 1 iterations
##
##   Estimator                                         ML
##   Optimization method                           NLMINB
##   Number of model parameters                         7
##
##   Number of observations                            93
##
## Model Test User Model:
##
##   Test statistic                                 5.137
##   Degrees of freedom                                 2
##   P-value (Chi-square)                           0.077
##
## Parameter Estimates:
##
##   Standard errors                             Standard
##   Information                                 Expected
##   Information saturated (h1) model          Structured
##
## Regressions:
##                    Estimate  Std.Err  z-value  P(>|z|)
##   min_T_winter ~
##     snow              0.793    0.063   12.544    0.000
##   soil_moist ~

##     snow                0.443    0.093    4.770    0.000
##   Carex.bigelowii ~
##     min_T_winter        0.339    0.103    3.278    0.001
##     soil_moist          0.040    0.103    0.386    0.700
##
## Variances:
##                    Estimate  Std.Err  z-value  P(>|z|)
##    .min_T_winter       0.367    0.054    6.819    0.000
##    .soil_moist         0.795    0.117    6.819    0.000
##    .Carex.bigelowi     0.862    0.126    6.819    0.000
```